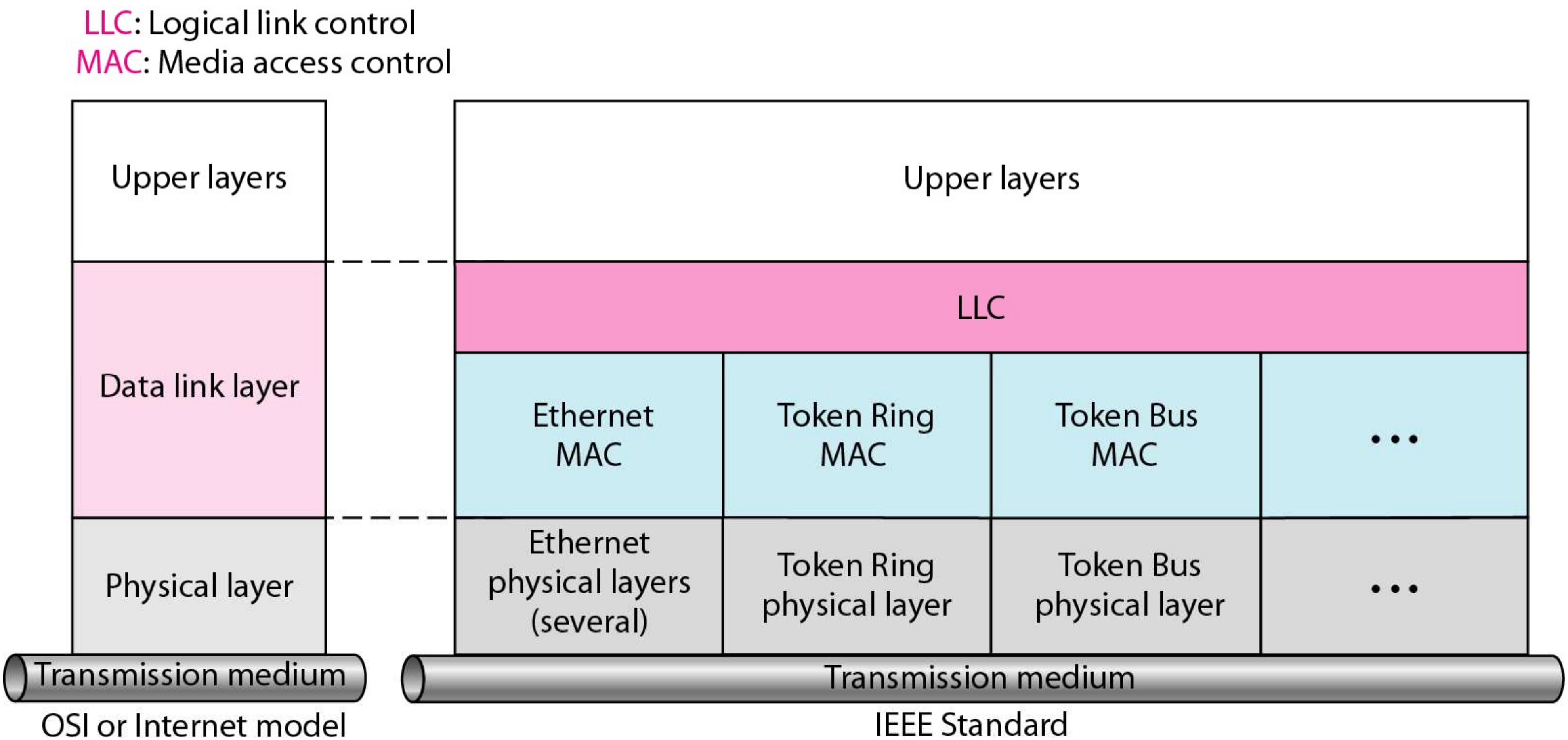
Chapter 6
Data Communications and Networking

Standardized Networks

IEEE standard for LANs

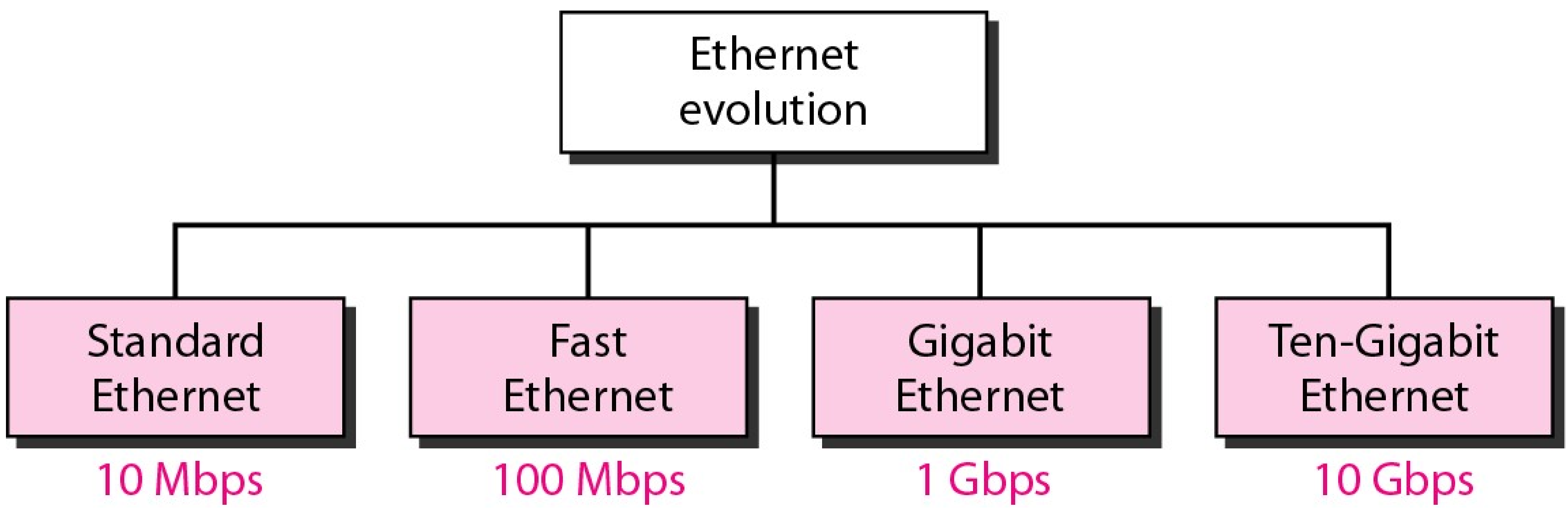
- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers.
- Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.

IEEE standard for LANs



Ethernet (IEEE 802.3)

- The original Ethernet was created in 1976 at Xerox's Palo Alto Research Center (PARC). Since then, it has gone through several generations.
- We briefly discuss the Standard Ethernet in this section.

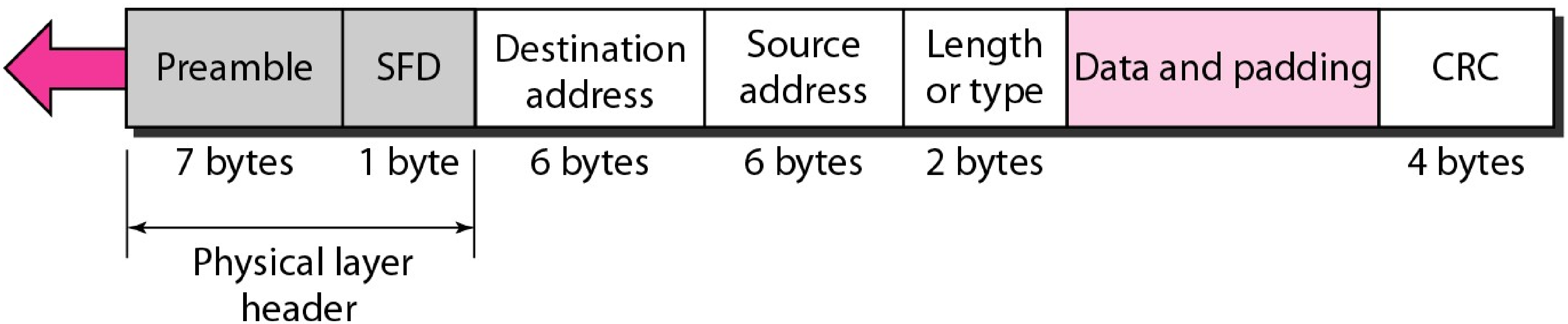


Ethernet (IEEE 802.3)

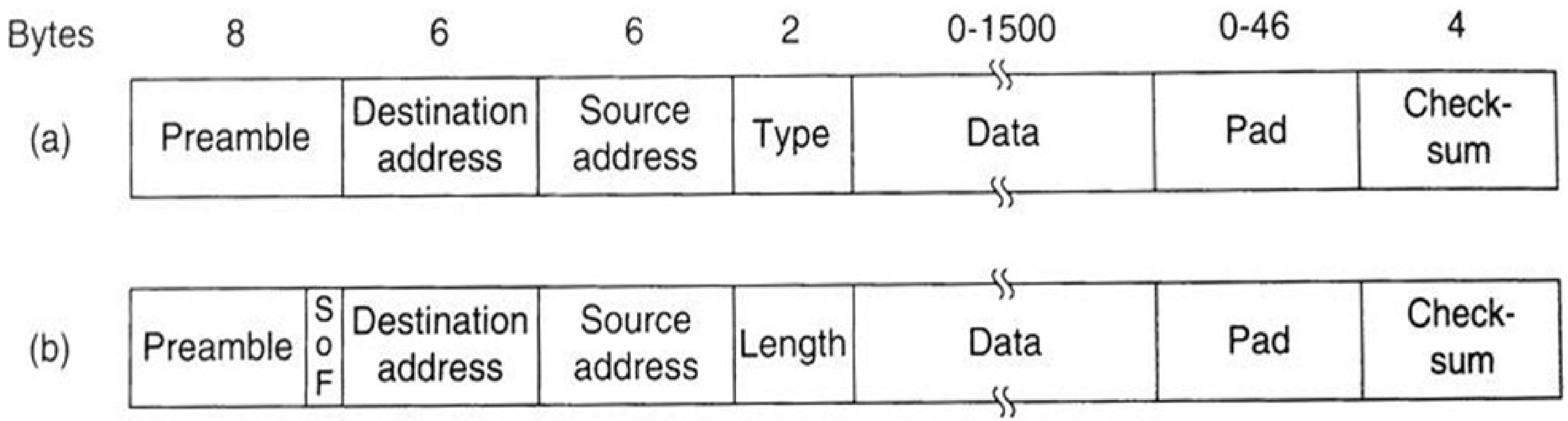
- 802.3 MAC frame

Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)



Frame Format

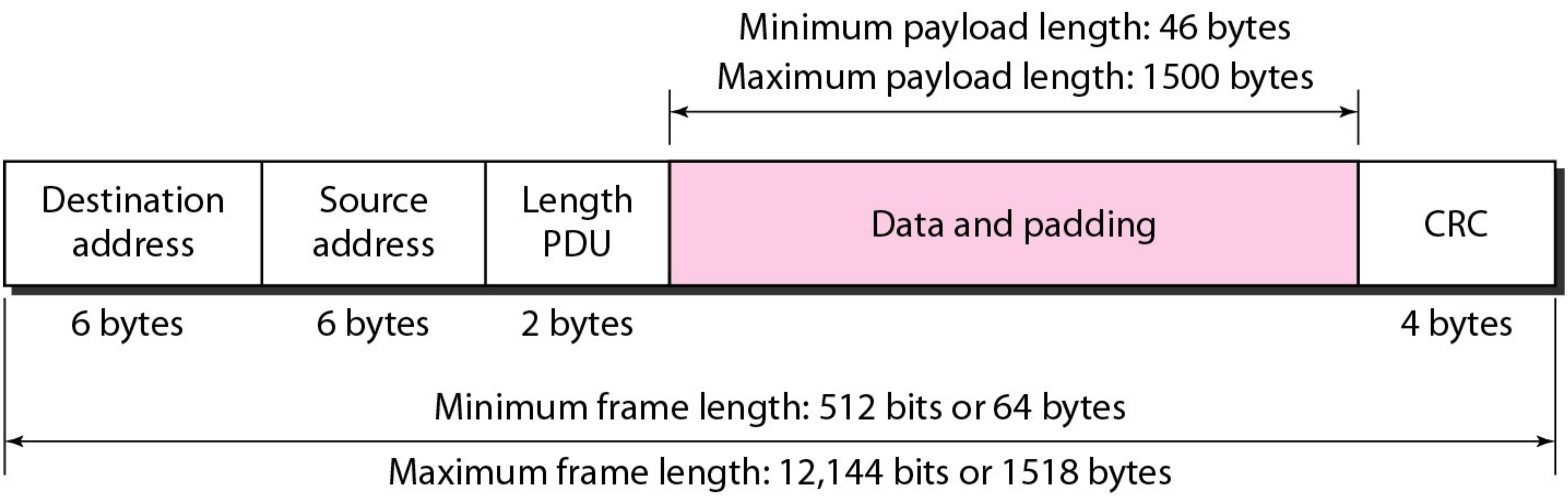


Frame formats. (a) DIX Ethernet. (b) IEEE 802.3.

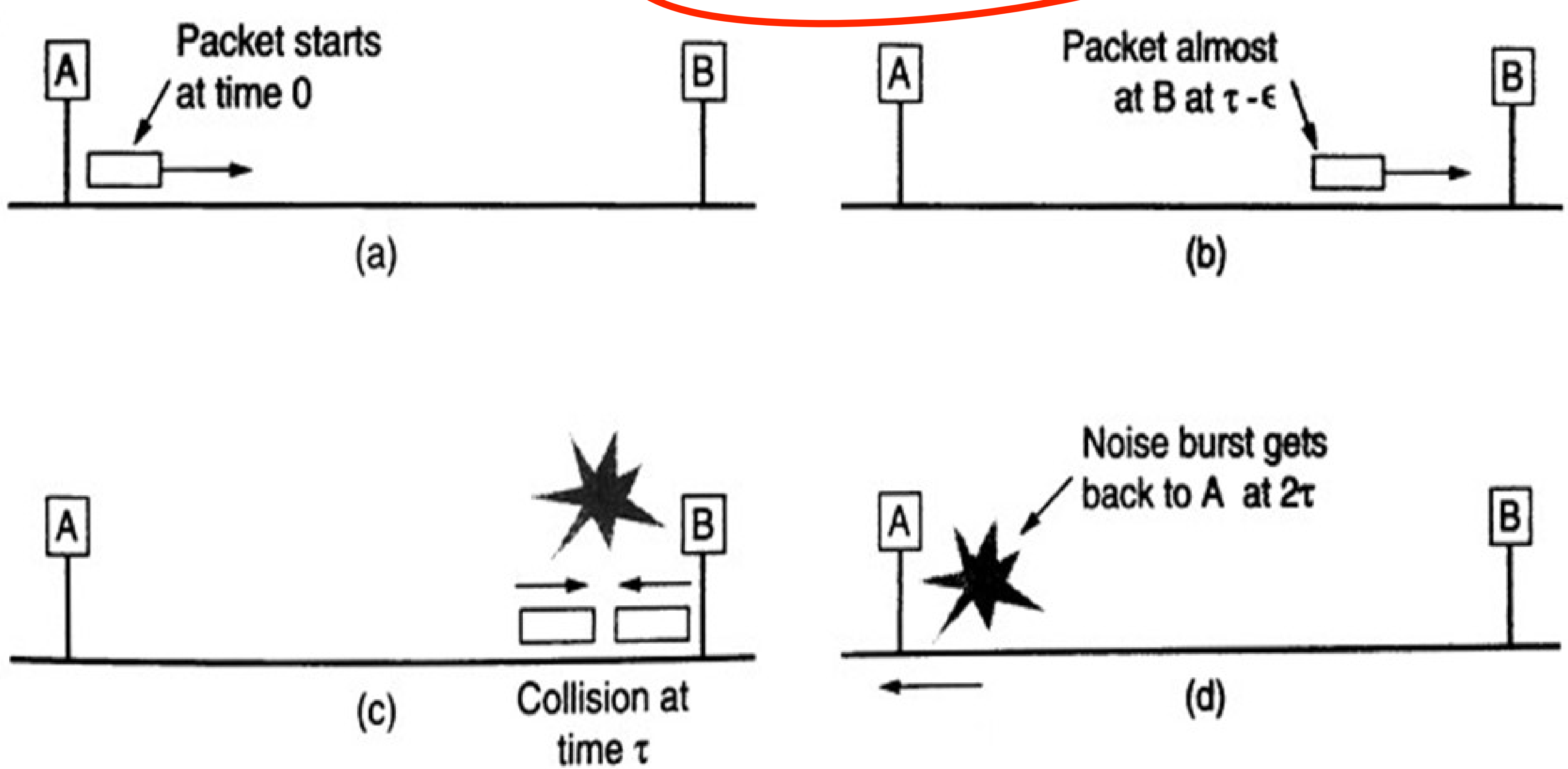
- Preamble: For synchronization with bit pattern 10101010.
- Destination and Source Addresses: 6 bytes.
- Type: this filed specifies which process to give the frame to
- Data: up to 1500 bytes
- Pad: Ethernet requires that valid frame must be at least 64 bytes long. The pad field is used to fill out the frame to the minimum size.

Ethernet (IEEE 802.3)

- Minimum and maximum lengths



Determination of the Minimum Frame Size



Collision detection can take as long as 2τ.

- How to determine the minimum frame size?

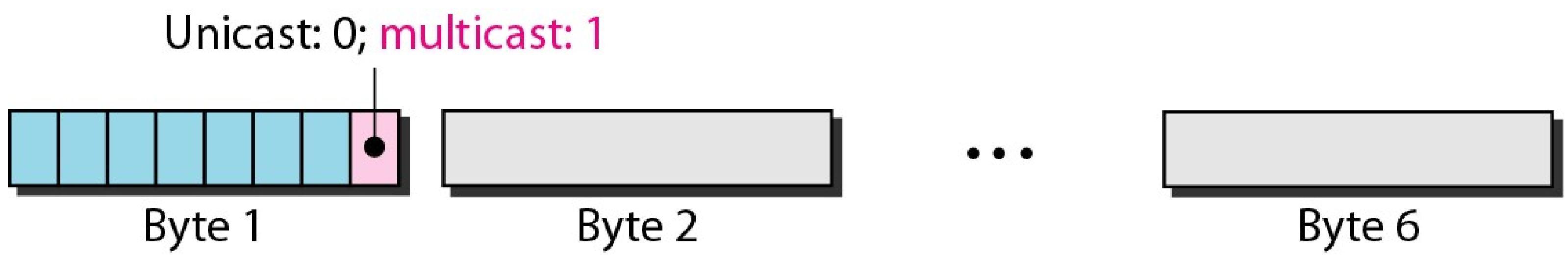
Ethernet (IEEE 802.3)

- Example of an Ethernet address in hexadecimal notation

$$\underbrace{\texttt{06 : 01 : 02 : 01 : 2C : 4B}}_{\text{6 bytes = 12 hex digits = 48 bits}}$$

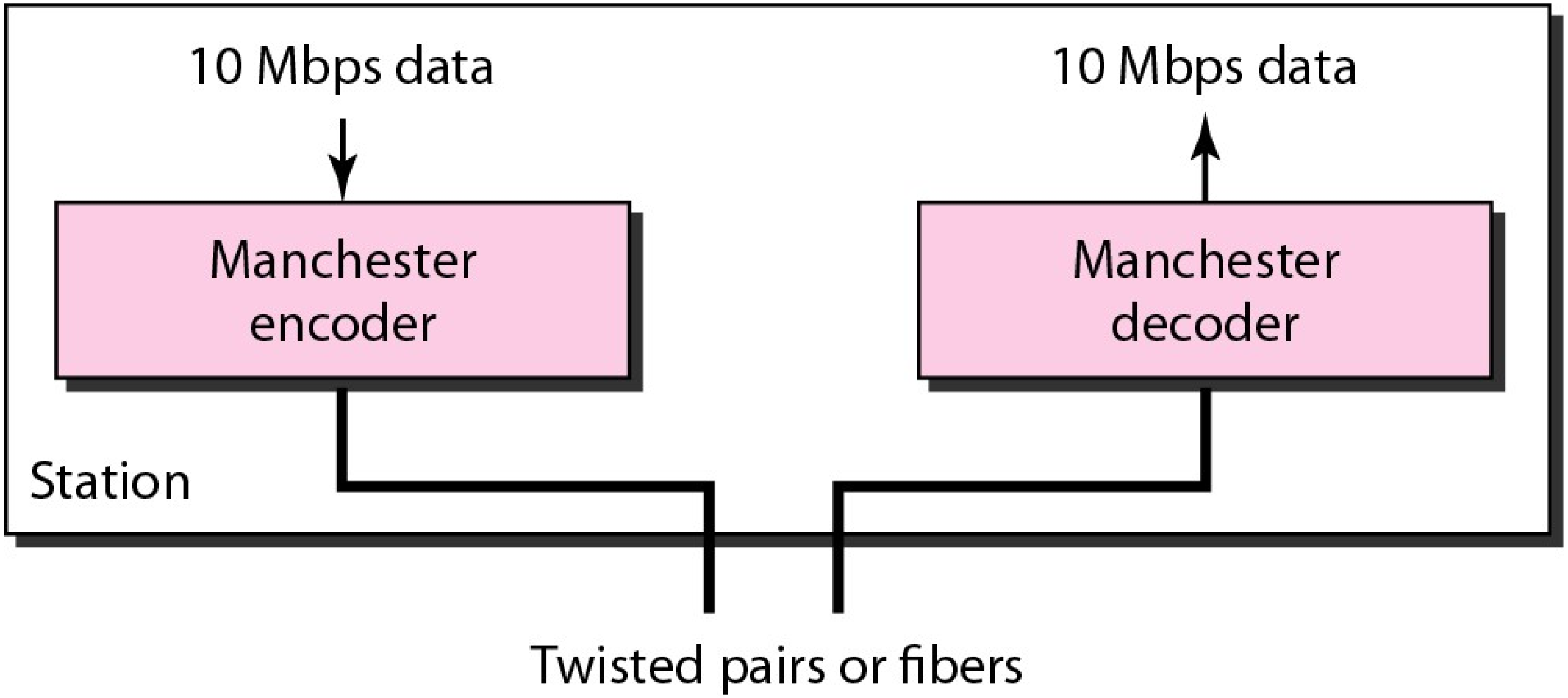
Ethernet (IEEE 802.3)

- Unicast and multicast addresses

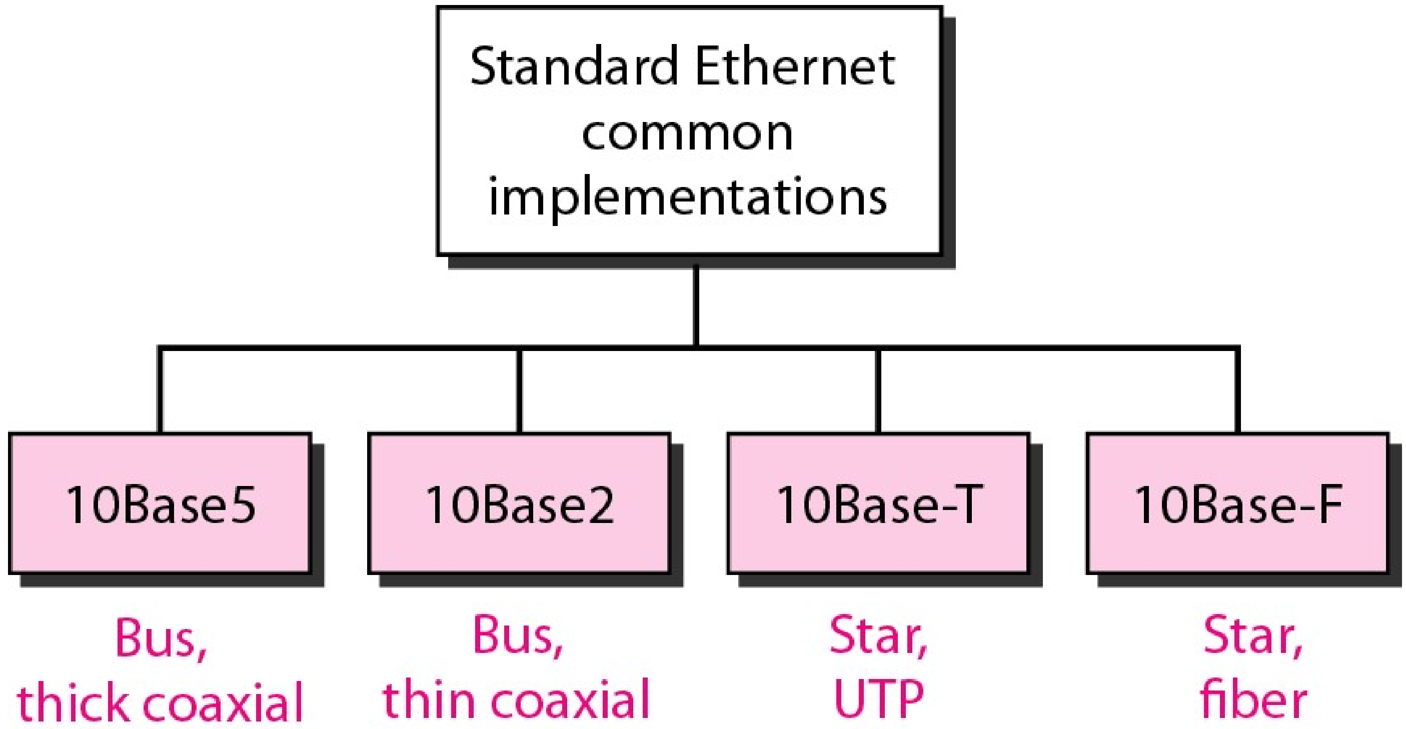


- The least significant bit of the first byte defines the type of address. If the bit is 0, the address is unicast; otherwise, it is multicast.
- The broadcast destination address is a special case of the multicast address in which all bits are 1s.

Encoding in a Standard Ethernet Implementation



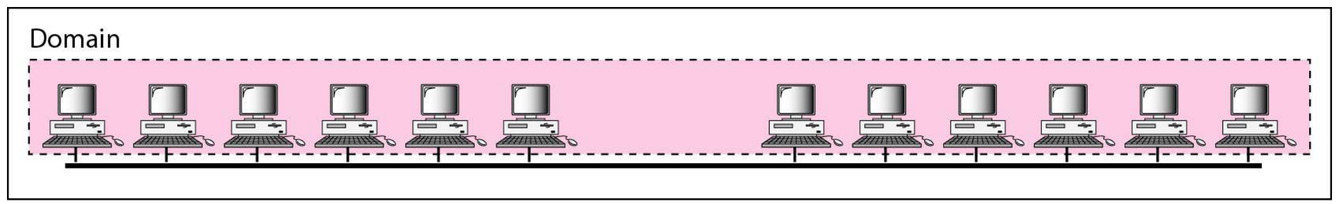
Categories of Standard Ethernet



Characteristics	*10Base5*	*10Base2*	*10Base-T*	*10Base-F*
Media	Thick coaxial cable	Thin coaxial cable	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m
Line encoding	Manchester	Manchester	Manchester	Manchester

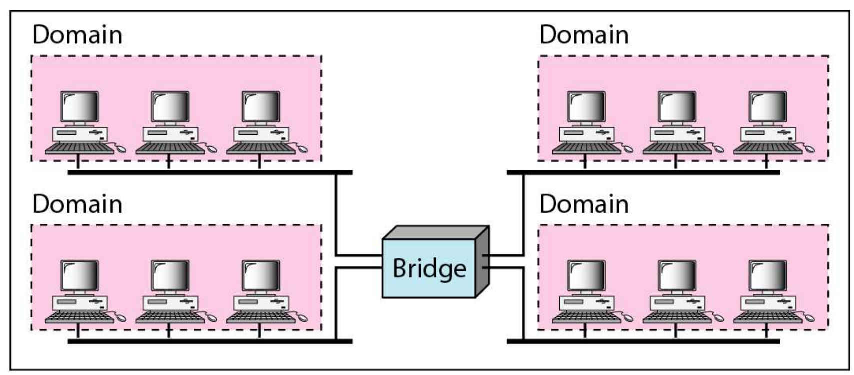
Changes in The Standard

- The 10-Mbps Standard Ethernet has gone through several changes before moving to the higher data rates.
- These changes actually opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs.



a. Without bridging

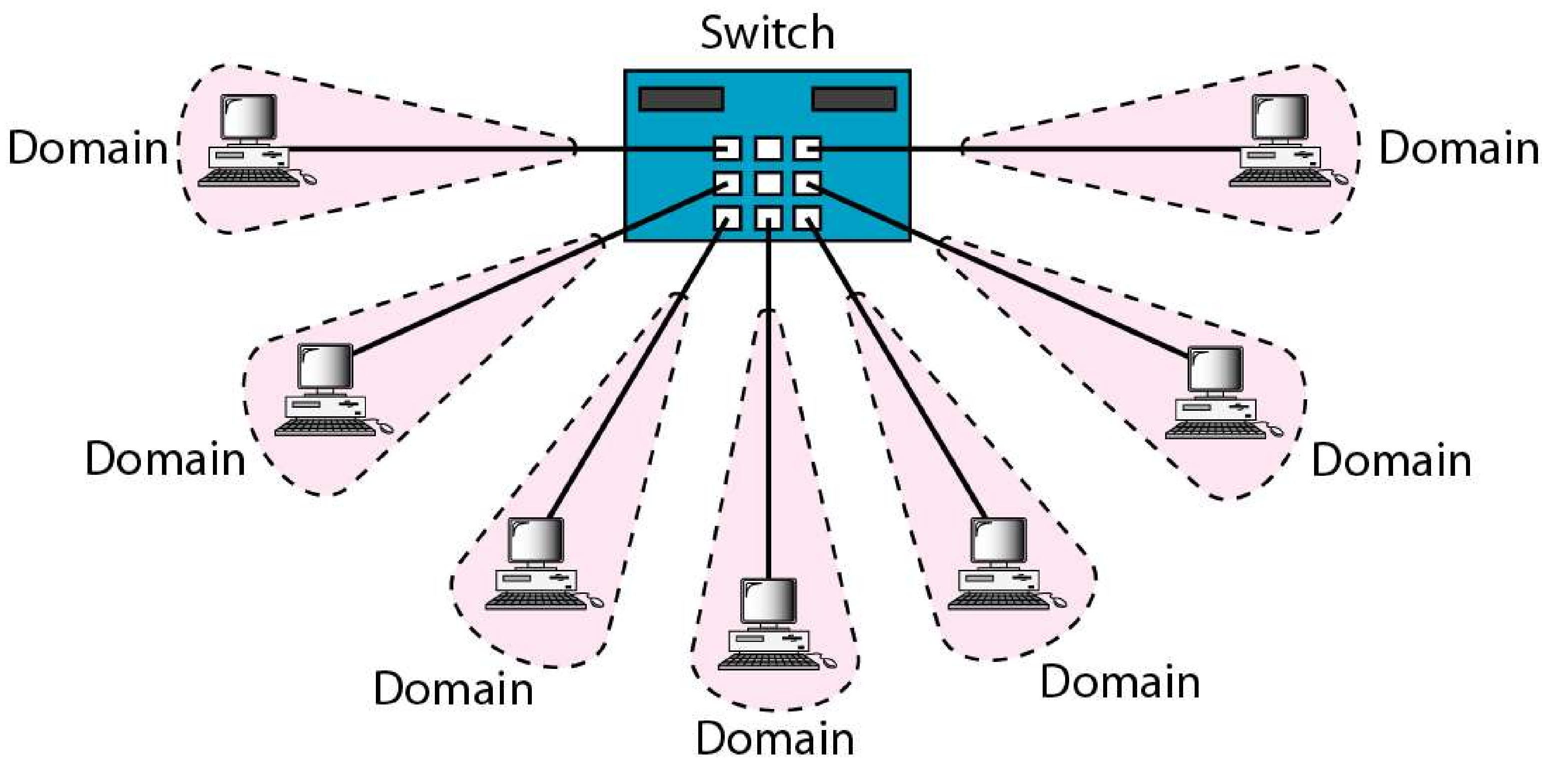
- Bridged network



b. With bridging

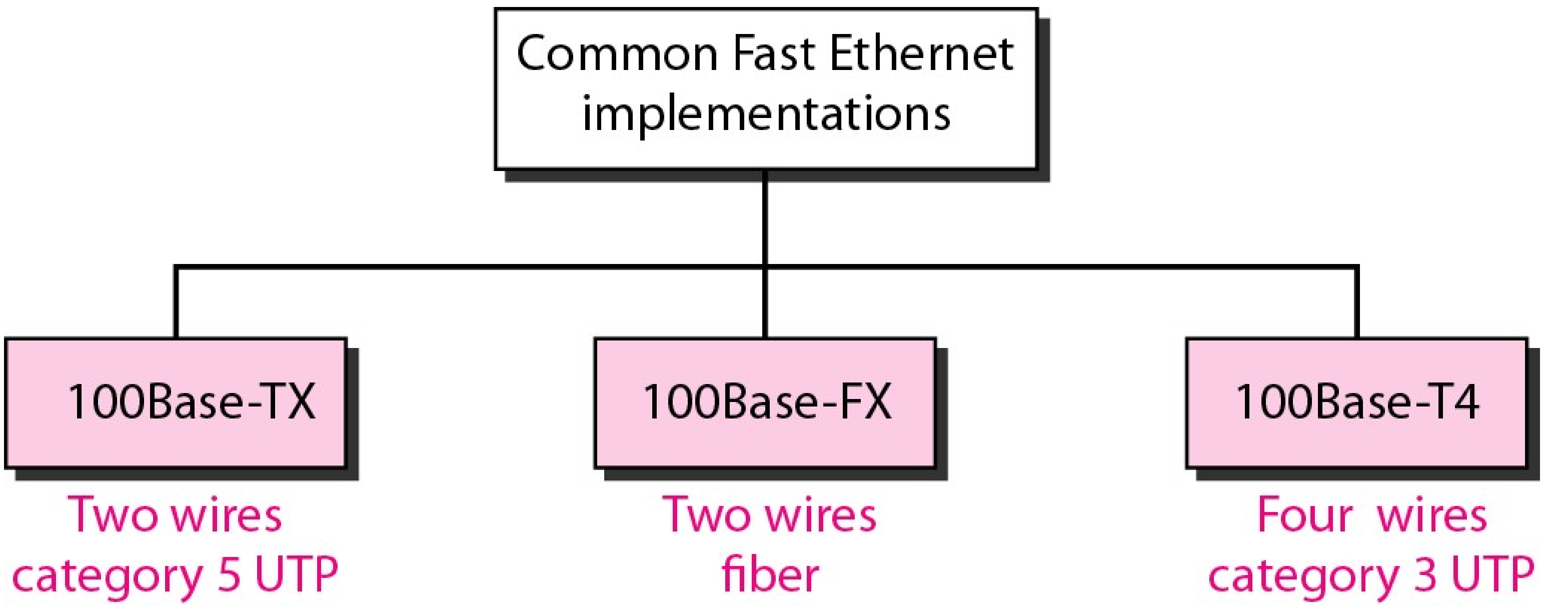
Changes in The Standard

- Switched Ethernet

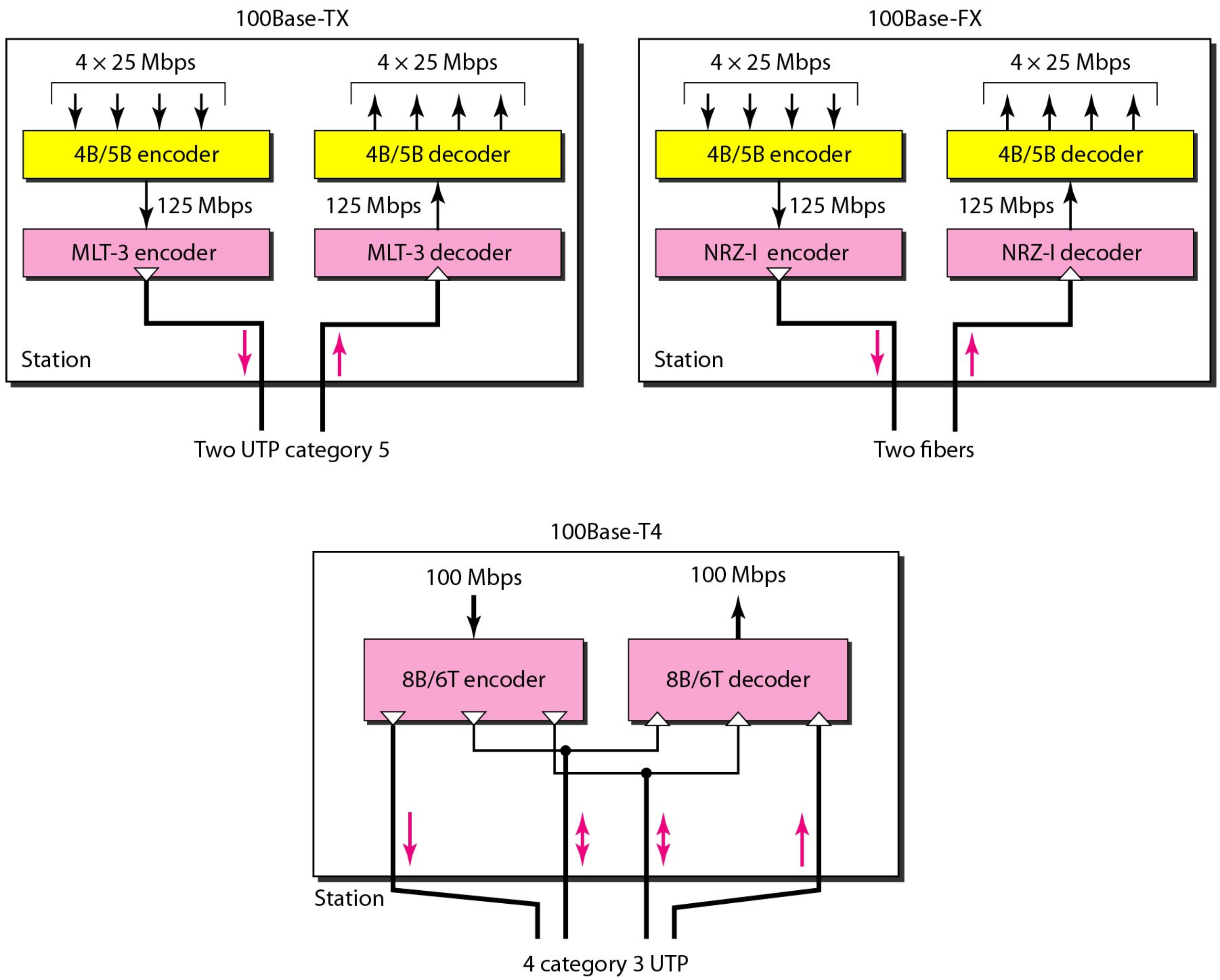


Fast Ethernet

- Fast Ethernet was designed to compete with LAN protocols such as FDDI or Fiber Channel.
- IEEE created Fast Ethernet under the name 802.3u.
- Fast Ethernet is backward-compatible with Standard Ethernet, but it can transmit data 10 times faster at a rate of 100 Mbps.



Encoding for Fast Ethernet implementation

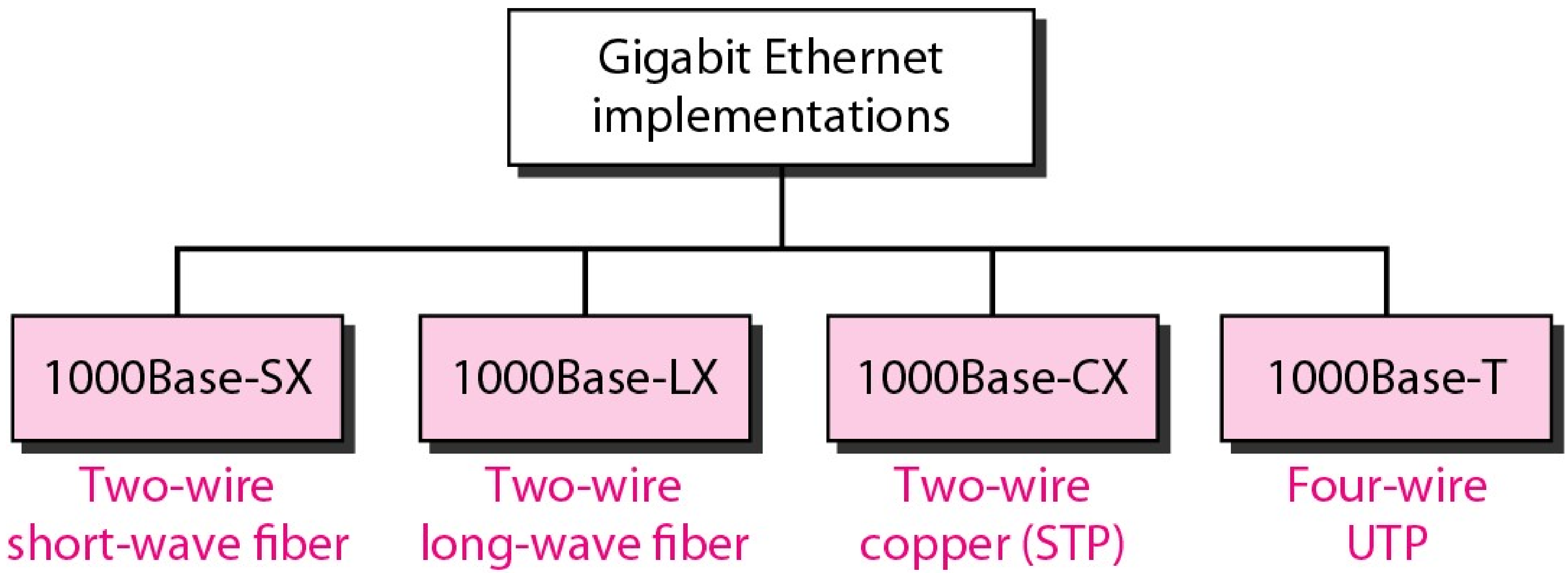


Summary of Fast Ethernet Implementations

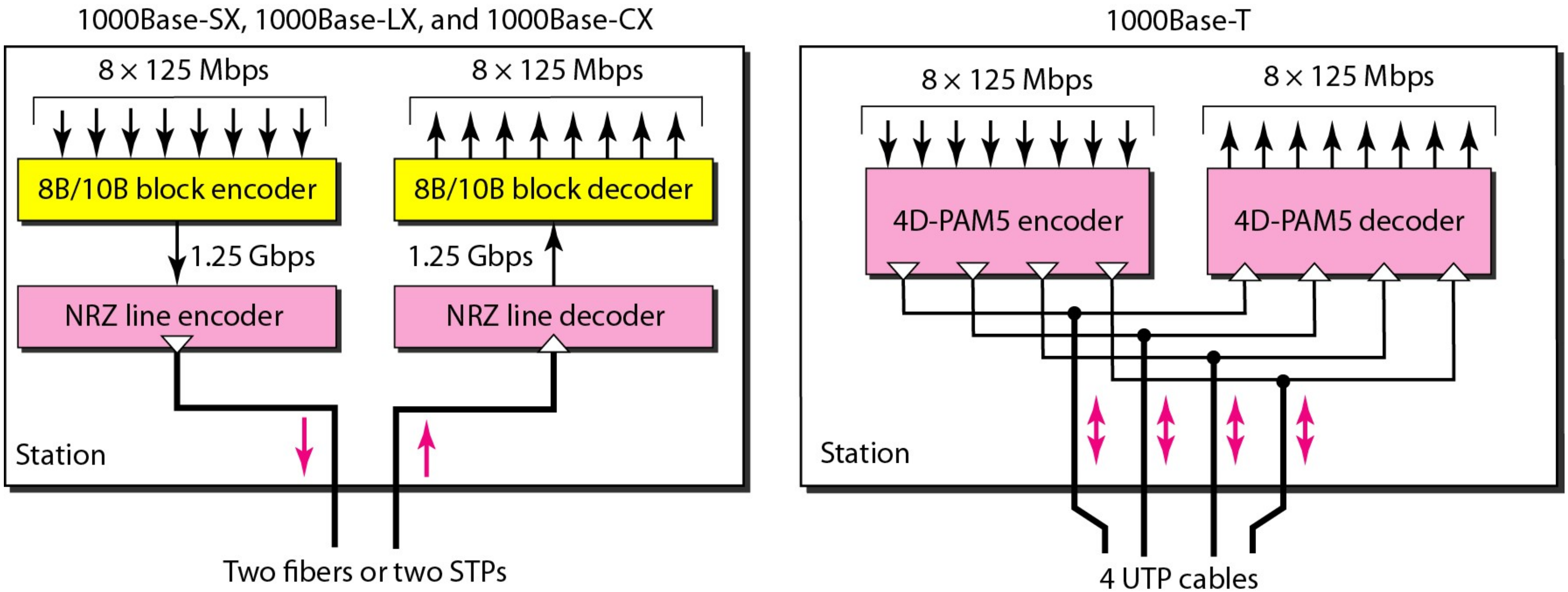
Characteristics	*100Base-TX*	*100Base-FX*	*100Base-T4*
Media	Cat 5 UTP or STP	Fiber	Cat 4 UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m
Block encoding	4B/5B	4B/5B	
Line encoding	MLT-3	NRZ-I	8B/6T

Gigabit Ethernet

- The need for an even higher data rate resulted in the design of the Gigabit Ethernet protocol (1000 Mbps).
- The IEEE committee calls the standard 802.3z.



Encoding in Gigabit Ethernet Implementations



Summary of Gigabit Ethernet Implementations

Characteristics	*1000Base-SX*	*1000Base-LX*	*1000Base-CX*	*1000Base-T*
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m
Block encoding	8B/10B	8B/10B	8B/10B	
Line encoding	NRZ	NRZ	NRZ	4D-PAM5

New Developments of Ethernet Standards

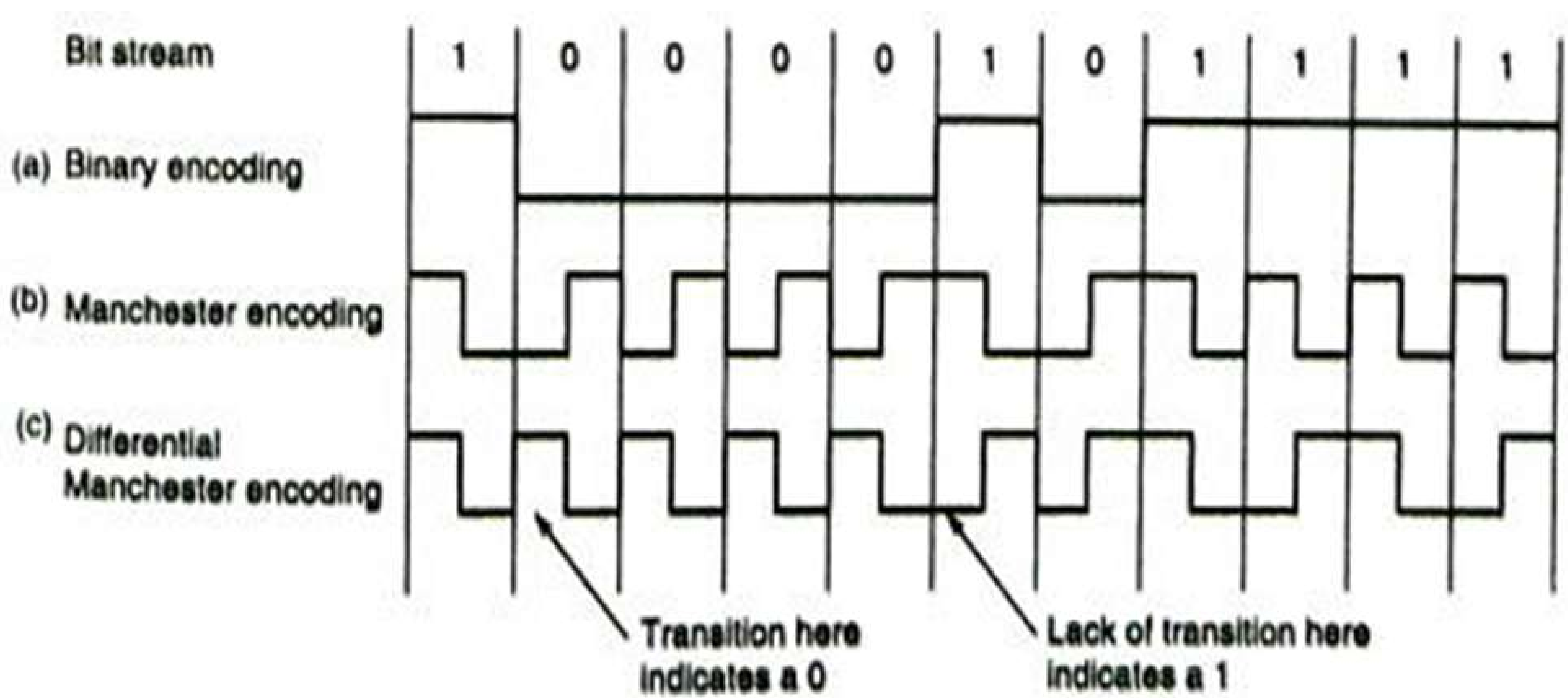
- Summary of 10-Gigabit Ethernet implementations

Characteristics	*10GBase-S*	*10GBase-L*	*10GBase-E*
Media	Short-wave 850-nm multimode	Long-wave 1310-nm single mode	Extended 1550-mm single mode
Maximum length	300 m	10 km	40 km

- 40G & 100 G Ethernet: The IEEE 802.3ba, a new standard governing 40G & 100G Ethernet was finalized in June 2010.

Manchester Encoding

- Ethernet uses Manchester encoding to transmit data on the LAN.
- With Manchester encoding, each bit contains a transition; a 1 has a transition from up to down, whereas a 0 has a transition from down to up.



Access Method

- Each Ethernet's adapter runs the CSMA/CD protocol without explicit coordination with the other adapters on the Ethernet.
  - The adapter obtains a network-layer PDU from its parent node, prepares an Ethernet frame, and put the frame in an adapter buffer.
  - The adapter senses the channel: if it is idle, the adapter starts to transmit the fame; if it is busy, the adapter waits until the channel is idle, then starts to transmit the frame.
  - While transmitting, the adapter monitors for the presence of signal energy coming from other adapters. If the adapter transmits the entire frame without collision detection, the transmission is successful.
  - If the adapter detects a collision while transmitting, it stops transmitting its frame and instead transmits a 48-bit jam signal.
  - After aborting, the adapter waits a random time (using Binary Exponential Back-off algorithm) and retry.

Binary Exponential Back-off Algorithm

- When a collision occurs, the current transmission is aborted and the adapter waits a random time to retry.
- Binary Exponential Back-off Algorithm is used to choose a random time dynamically.
  - After the first collision, each station waits either 0 or 1 slot time before trying again. If each one picks the same random number, they collide again.
  - After the second collision, each one picks either 0, 1, 2, or 3 at random and waits that number of slot times.
  - In general, after *i* collisions, a random number between 0 and $2^i - 1$ is chosen, and the adapter waits that number of slot times. However, after ten collisions have been reached, the randomization interval is frozen at a maximum of ($2^{10} - 1$) slots.
  - After 16 collisions, the controller gives up and reports failure back to the computer.

Token Ring (IEEE 802.5)

- Token-passing networks move a small frame, called a token, around the network
- Possession of the token grants the right to transmit.
- If a node receiving the token has no information to send, it passes the token to the next end station
- If a station possessing the token does have information to transmit
  - It seizes the token, alters 1 bit of the token (which turns the token into a start-of-frame sequence),
  - Appends the information that it wants to transmit, and sends this information to the next station on the ring.
- Each station can hold the token for a maximum period of time

Token Ring (IEEE 802.5)

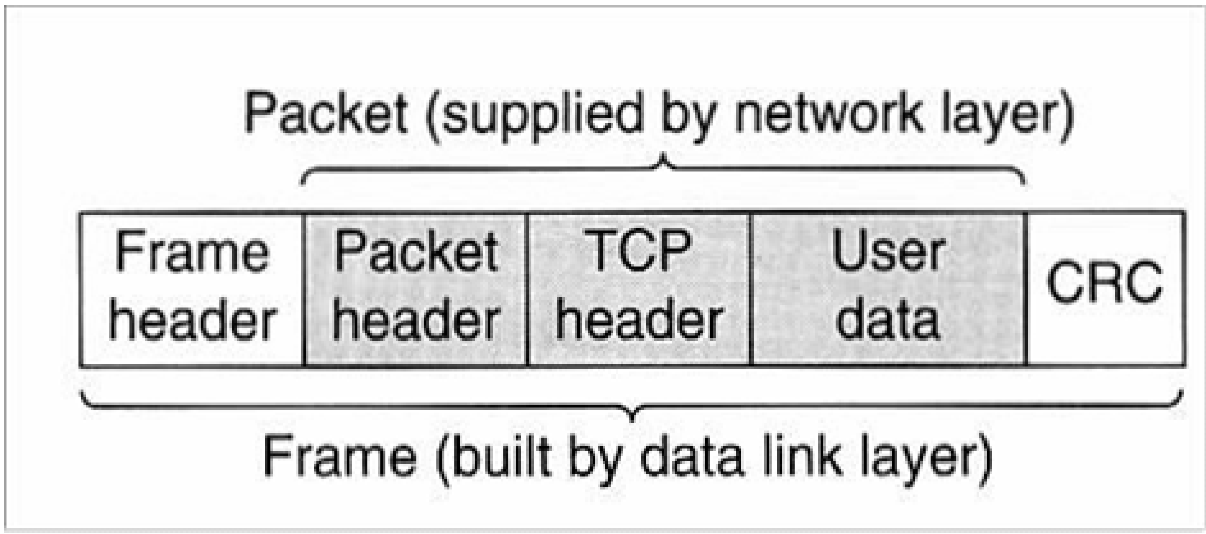
- Priority System
  - Token Ring frames have two fields that control priority: the priority field and the reservation field.
  - Only stations with a priority equal to or higher than the priority value contained in a token can seize that token.
  - Only stations with a priority value higher than that of the transmitting station can reserve the token for the next pass around the network.
- Fault-Management Mechanisms
  - One station in the Token Ring network is selected to be the active monitor.
  - This station acts as a centralized source of timing information for other ring stations and performs a variety of ring-maintenance functions.
- Frame format: Support two basic frame types: tokens and data/command frames.

Hub, Bridge, Router, and Switch

- Hub: hub is a repeater with some additional network management functionality (such as performance or accounting management)
- Bridge: bridge operates on Ethernet frames and thus a layer-2 device. It does the following two functions:
  - Filtering: determine whether a frame should be forwarded to some interface or should just be dropped.
  - Forwarding: determine the interfaces to which a frame should be directed.

Hub, Bridge, Router, and Switch

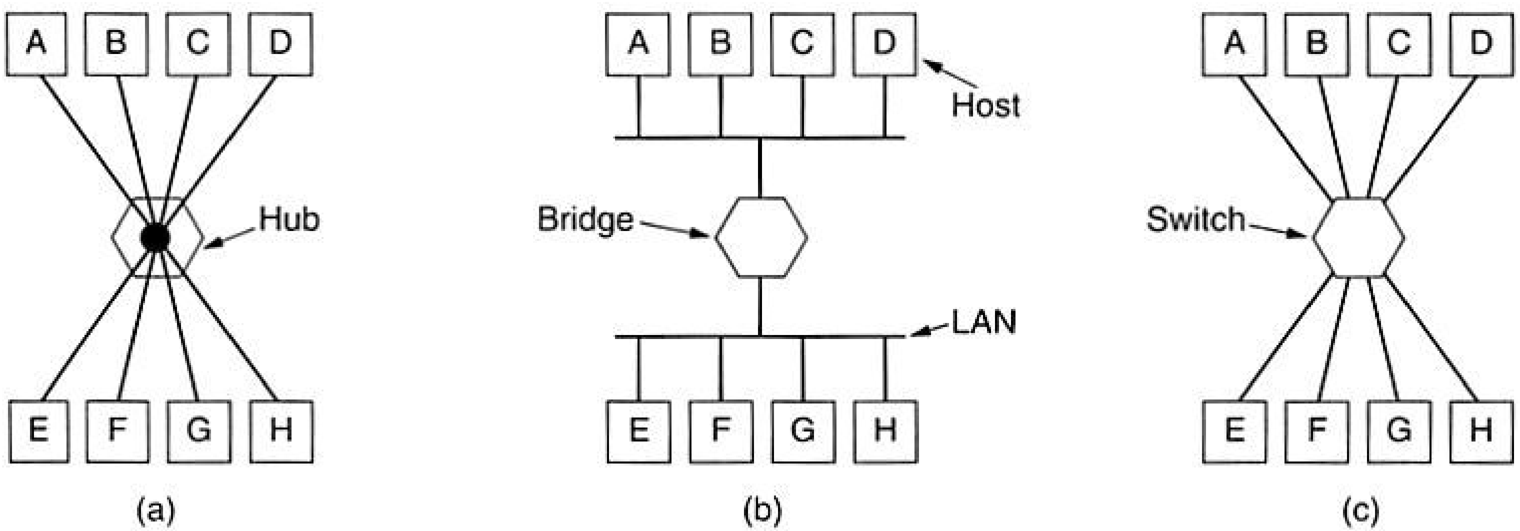
- Router: router is a store-and-forward packet switch that forward packets using network-layer addresses (layer-3).
- Although a bridge is also a store- and-forward packet switch, it forwards packets using LAN addresses.



- As a network administrator, how to choose between bridge and router?

Hub, Bridge, Router, and Switch

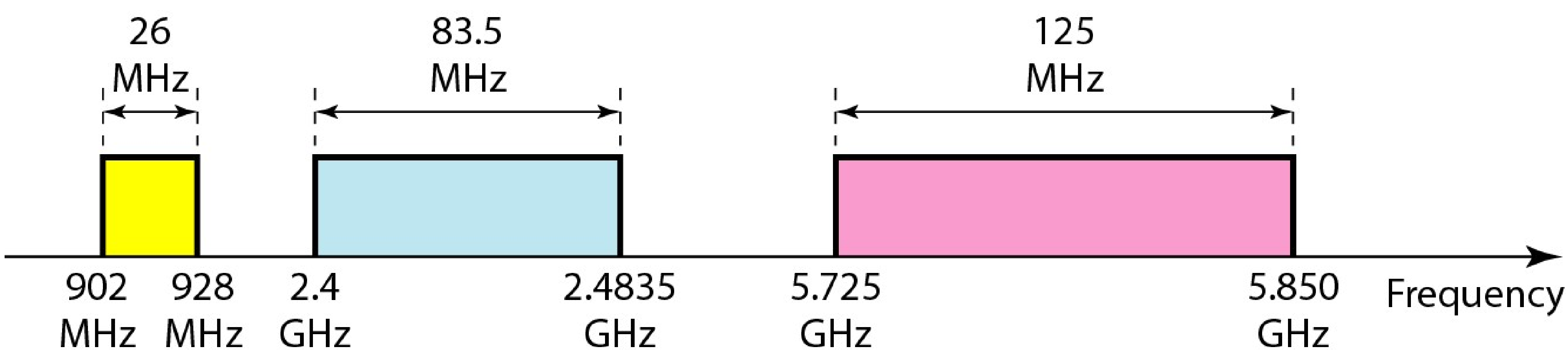
- Switch: switch is in essence a high-performance multi-interface bridge. The difference between a bridge and switch:
- Bridges usually two or four interfaces, whereas switches have dozens of interfaces (e.g., 24 ports).
- Switches are usually used to connect individual computers, and operate in a full-duplex mode.



(a) A hub. (b) A bridge. (c) A switch.

Wireless LANs (IEEE 802.11)

- IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data link layers.
- Industrial, scientific, and medical (ISM) band



Wireless LANs (IEEE 802.11)

- Currently, the most popular LAN standards are 802.11b (11 Mbps), 802.11g (54 Mbps) and 802.11n (600Mbps).
- Channel Allocation in Different Countries

Channel	Frequency (MHz)	North America	Japan	Most of world
1	2412	Yes	Yes	Yes
2	2417	Yes	Yes	Yes
3	2422	Yes	Yes	Yes
4	2427	Yes	Yes	Yes
5	2432	Yes	Yes	Yes
6	2437	Yes	Yes	Yes
7	2442	Yes	Yes	Yes
8	2447	Yes	Yes	Yes
9	2452	Yes	Yes	Yes
10	2457	Yes	Yes	Yes
11	2462	Yes	Yes	Yes
12	2467	No	Yes	Yes
13	2472	No	Yes	Yes
14	2484	No	**.11b** only	No

Wireless LANs (IEEE 802.11)

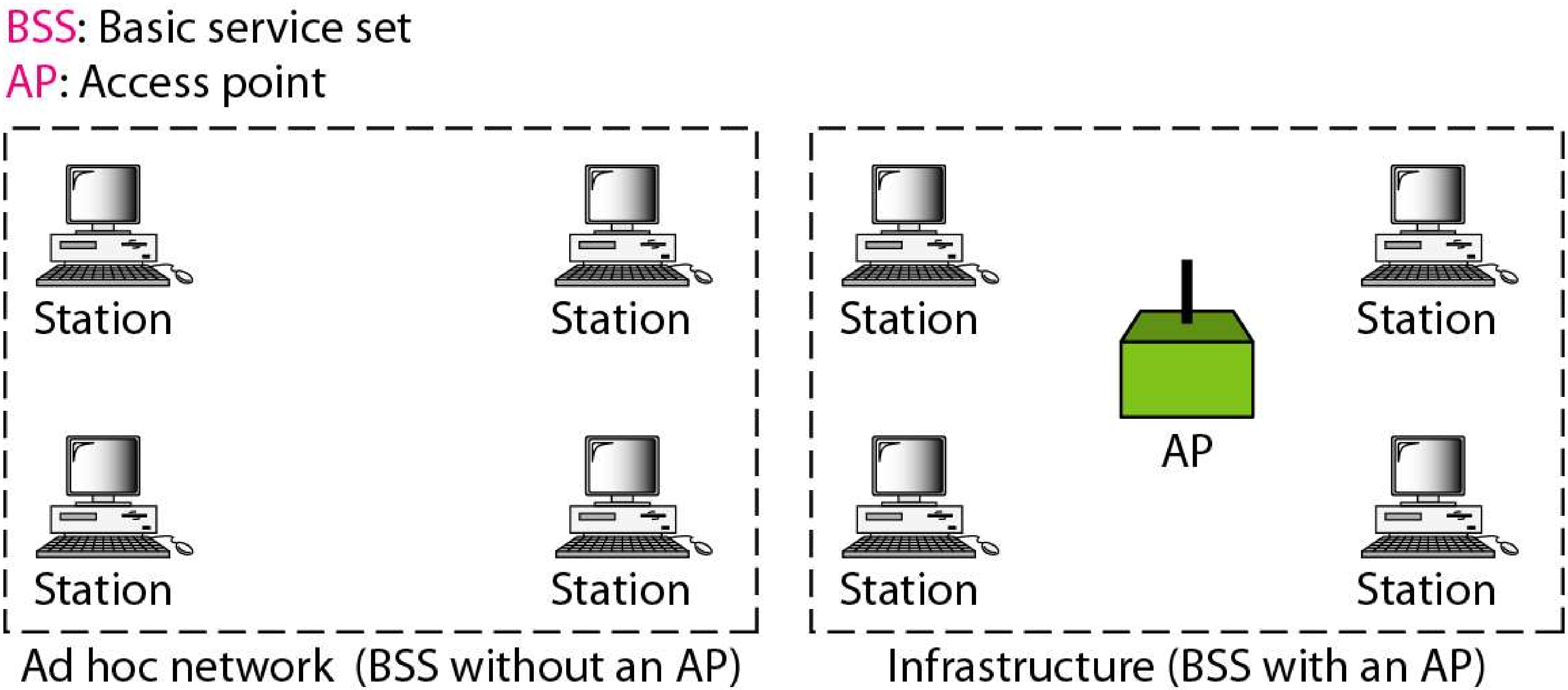
802.11 network standards v · d · e [hide]										
802.11 Protocol	Release[7]	Freq. (GHz)	Bandwidth (MHz)	Data rate per stream (Mbit/s)[8]	Allowable MIMO streams	Modulation	Approximate indoor range[citation needed]		Approximate Outdoor range[citation needed]	
							(m)	(ft)	(m)	(ft)
–	Jun 1997	2.4	20	1, 2	1	DSSS, FHSS	20	66	100	330
a	Sep 1999	5	20	6, 9, 12, 18, 24, 36, 48, 54	1	OFDM	35	115	120	390
		3.7[y]					--	--	5,000	16,000[y]
b	Sep 1999	2.4	20	5.5, 11	1	DSSS	38	125	140	460
g	Jun 2003	2.4	20	6, 9, 12, 18, 24, 36, 48, 54	1	OFDM, DSSS	38	125	140	460
n	Oct 2009	2.4/5	20	7.2, 14.4, 21.7, 28.9, 43.3, 57.8, 65, 72.2[z]	4	OFDM	70	230	250	820[9]
			40	15, 30, 45, 60, 90, 120, 135, 150[z]			70	230	250	820[9]

- [y] IEEE 802.11y-2008 extended operation of 802.11a to the licensed 3.7 GHz band. Increased power limits allow a range up to 5000m. As of 2009, it is only being licensed in the United States by the FCC.
- [z] Assumes Short Guard interval (SGI) enabled, otherwise reduce each data rate by 10%.

(Source: wikipedia.org)

Wireless LANs (IEEE 802.11)

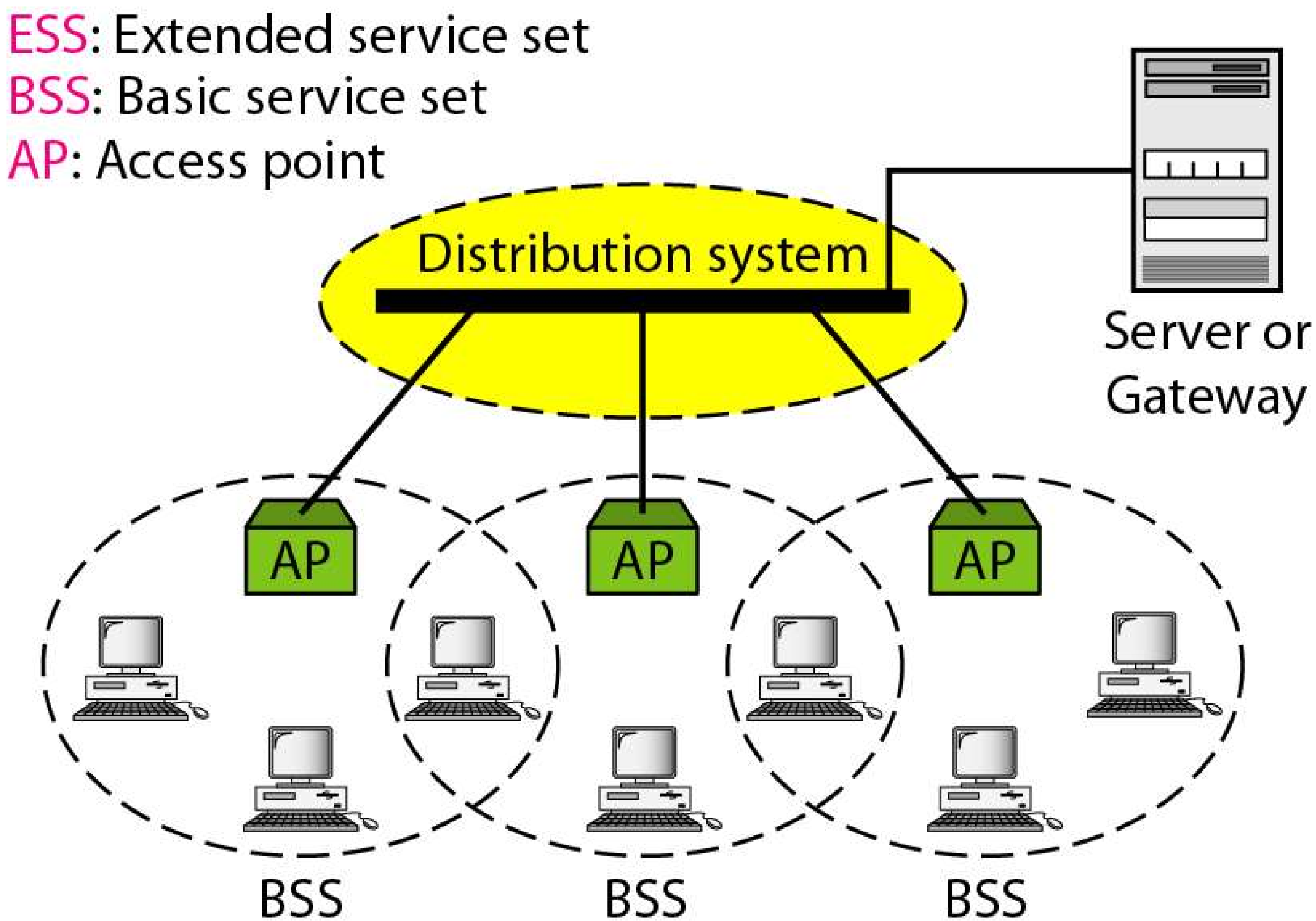
- Basic service sets (BSSs)



- A BSS without an AP is called an ad hoc network; a BSS with an AP is called an infrastructure network.

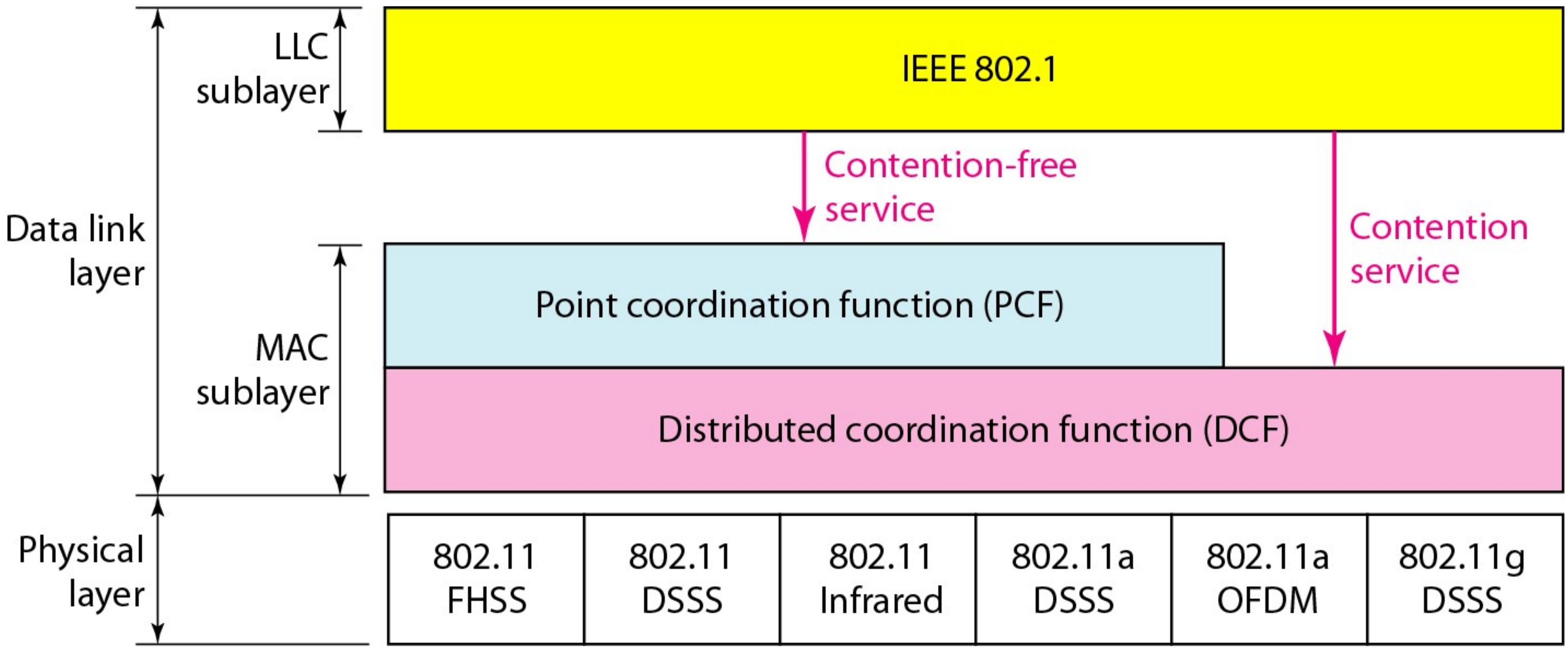
Wireless LANs (IEEE 802.11)

- Extended service sets (ESSs)



Wireless LANs (IEEE 802.11)

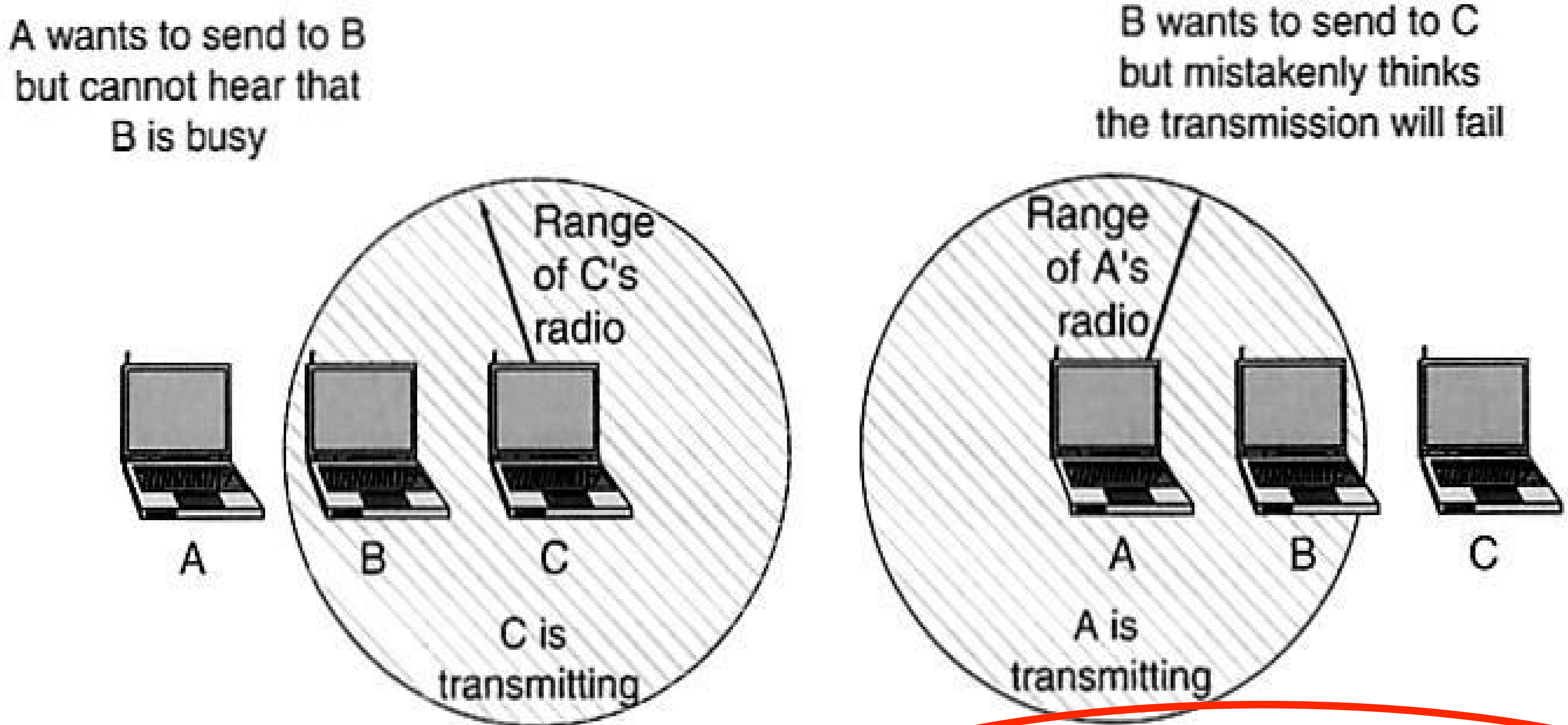
- MAC layers in IEEE 802.11 standard



Wireless LANs (IEEE 802.11)

- MAC protocol (CSMA/CD) of Ethernet does not work in wireless LAN.
- Not all stations are within radio range of each other, transmissions going on in one part of a cell may not be received elsewhere in the same cell.
- Hidden station problem: for example, station *C* is transmitting to station *B*. If *A* senses the channel, it will not hear anything and falsely conclude that it may now start transmitting to *B*.
- Exposed station problem: for example, station *B* wants to send to *C* so it listens to the channel. When it hears a transmission, it falsely concludes that it may not send to *C*, even though *A* may be transmitting to *D*.

Wireless LANs (IEEE 802.11)



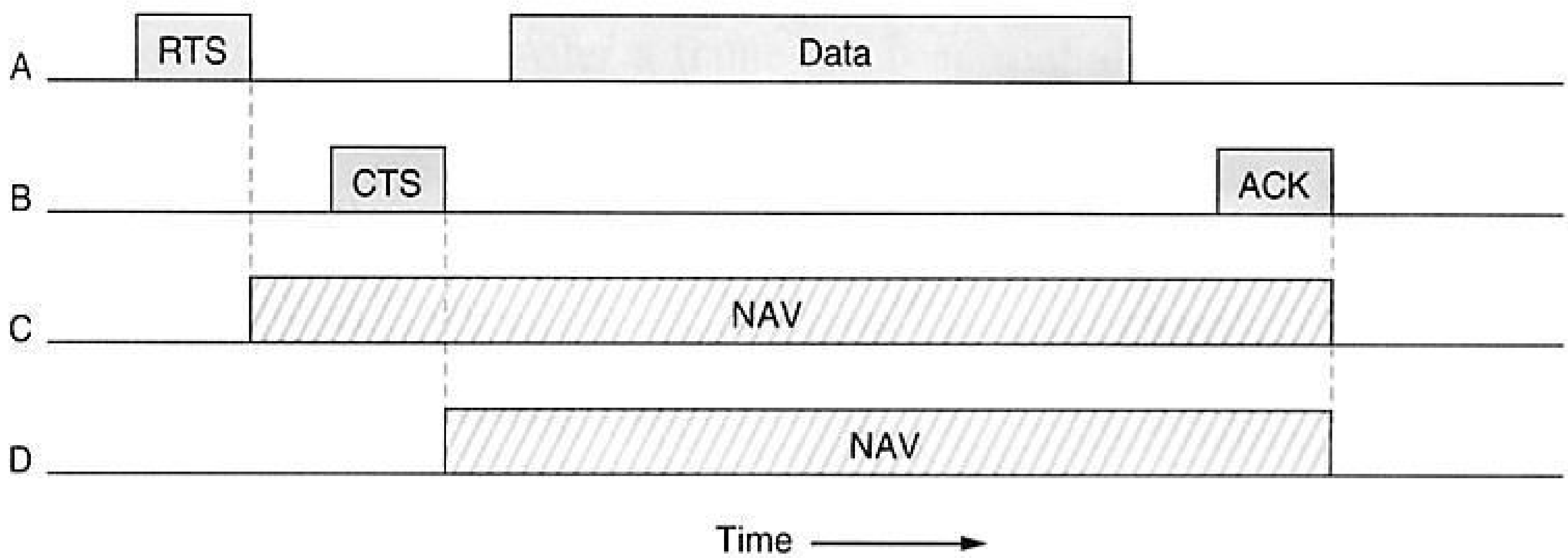
(a) The hidden station problem. (b) The exposed station problem.

Wireless LANs (IEEE 802.11)

- 802.11 supports CSMA/CA (CSMA with collision avoidance) for MAC protocol.
  - Suppose there are 4 stations. *A* wants to send to *B*. *C* is within range of *A*. *D* is within range of *B* but not within range of *A*.
  - *A* begins by sending an RTS (request to send) frame to *B* to request permission to send.
  - If *B* decides to grant permission, it replies a CTS (clear to send) frame.
  - Upon receipt of the CTS, *A* sends its frame and starts an ACK timer.
  - Upon the correct receipt of the data frame, *B* responds with an ACK frame, terminating the exchange.
  - For *C*, it is within range of *A*, it will receive the RTS frame. Then it asserts a kind of virtual channel busy for itself, indicated by **NAV** (network allocation vector).
  - For *D*, it does not hear the RTS, but it does hear the CTS, so it also asserts the NAV signal for itself.

Wireless LANs (IEEE 802.11)

- CSMA/CA



Wireless LANs (IEEE 802.11)

- PCF (point coordination function): the base station polls the other stations, asking them if they have any frames to send. → No collision.
- Mechanism: the base station broadcast a **beacon frame** periodically.
  - Ask other stations if they have any frame to send.
  - Invite new stations to sign up for polling service.

Bluetooth (IEEE 802.15)

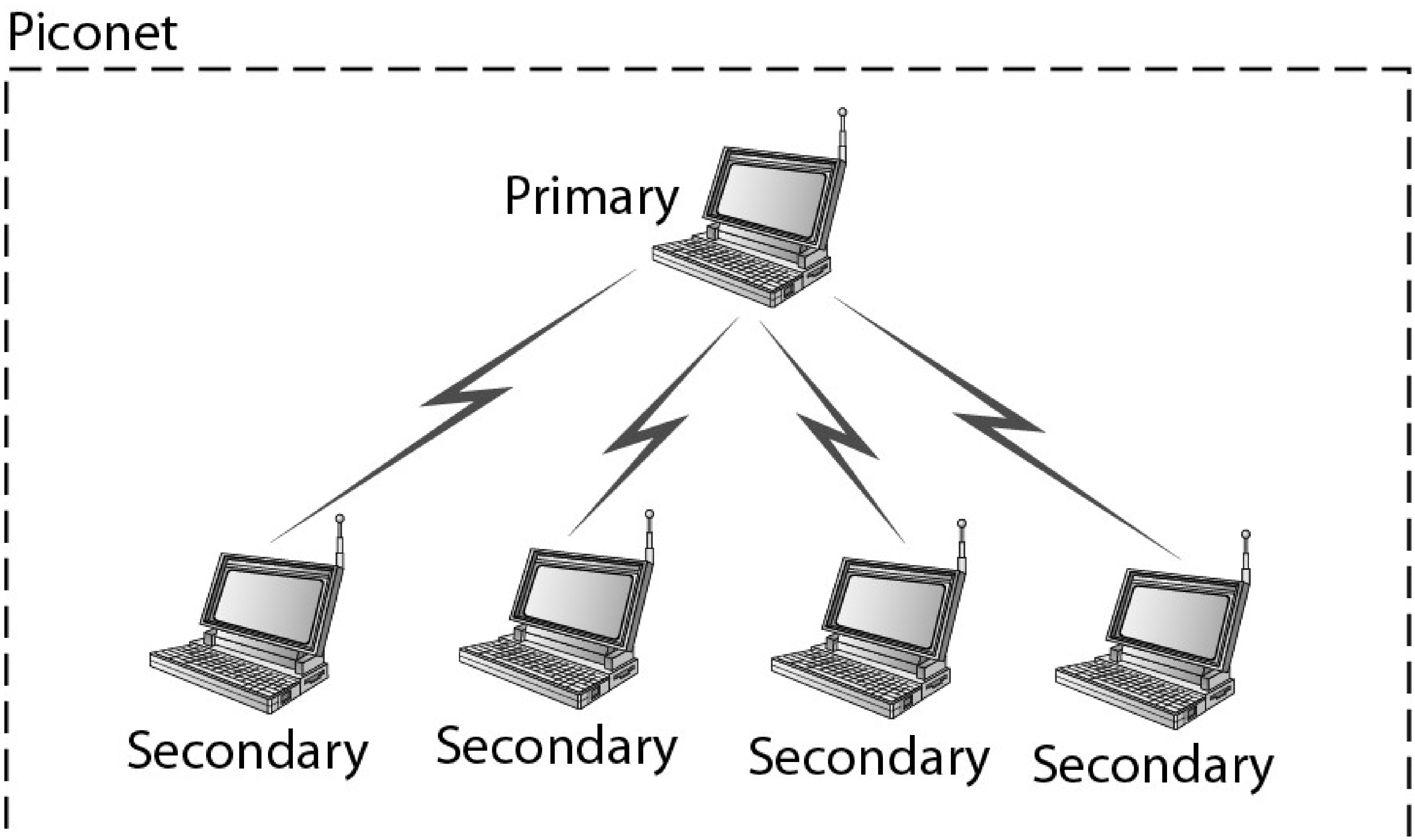
- Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, coffee makers, and so on.
- A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.
- It operates in the unlicensed 2.4 GHz spectrum.
- The basic unit of a Bluetooth system is a piconet, which consists of a master node and up to seven active slave nodes (and up to 255 parked nodes) with in a distance of 10 meters normally (could be up to 100 meters).

Bluetooth (IEEE 802.15)

- Multiple piconets can exists in the same room and can be connected via a bridge node.
- The slaves just do whatever the master tells them to do.
- The data-rate ranges from 1Mbit/sec to 24Mbit/sec (depending on the versions)
- Lower power, lower cost & data encryption

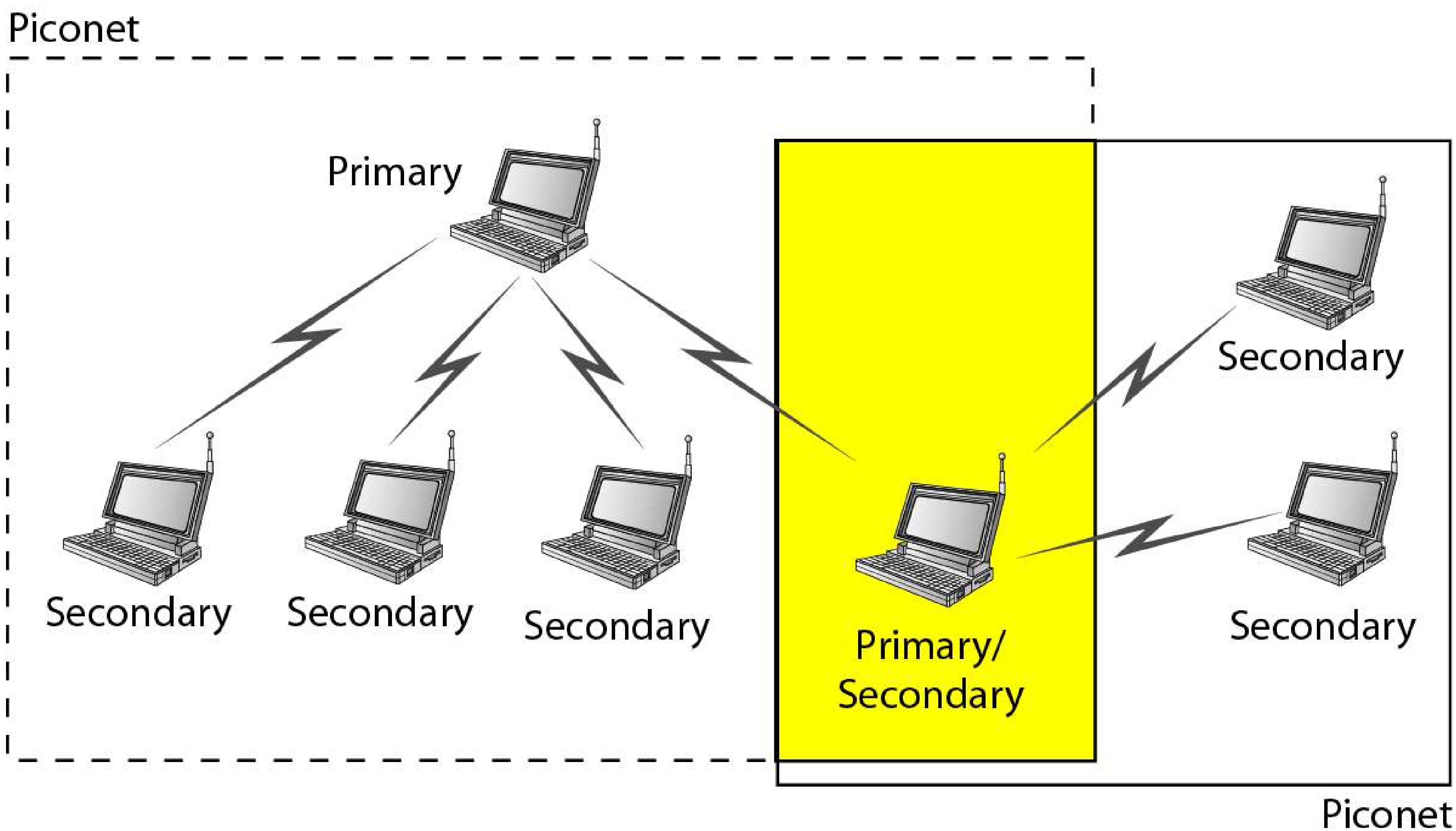
Bluetooth (IEEE 802.15)

- Piconet



Bluetooth (IEEE 802.15)

- Scatternet



Bluetooth (IEEE 802.15)

- Scatternet